High Density Single Port Register File SVT MVT Compiler
CLN22ULL-GL 22nm Process
256 Rows Per Bit Line, 0.192um^2 BitCell
128 Words X 128 Bits, Mux 2 Instance

Overview

The High Density Single Port Register File SVT MVT Compiler is optimized for speed and density. The memory is designed to take full advantage of the TSMC 22nm CLN22UL CMOS process.

The storage array is composed of six-transistor bit cells with fully static circuitry. The register file operates at a voltage of 0.88V and a junction temperature of -40.0°C.

Instance Settings

Parameter	Setting
Instance Name	rf128x128
Process	CLN22UL
Number of Words	128
Bits	128
Multiplexer Width	2
Multi-Vt selection	BASE
Frequency <MHz>	500
Activity Factor <%>	50
Word-Write Mask	on
Word Partition Size	1
Write through	off
Top Metal Layer	m5-m10
Power Type	otc
Redundancy	off
Redundant Columns	2
Redundant Rows	0
BIST MUXes	off
Soft Error Repair (SER)	none
Power Gating	off
Back Biasing	off
Retention	on
Extra Margin Adjustment	on
Advanced Test Features	off
Vmin Assist	on

Description

Register file access is synchronous and is triggered by the rising-edge of the clocks, clk.Input addresses, input data, write enables and chip enables are latched by the rising-edges of their respective clocks, respecting individual setup and hold times.

A read cycle is initiated in the register file if cen is low and gwen is high at the rising-edge of the clock, clk. The contents of the RAM location specified by the address, a, are driven on the data output bus, q.

Description (cont)

The register file is allowed to access non-existing physical addresses, but the outputs will be unknown.

The read address for any given memory cycle can be identical to the write address of the previous memory cycle with the read data being identical to the data that was written from the previous memory write cycle.

A write cycle is initiated if the write enable, gwen is low, and the chip enable, cen, is low at the rising edge of the clock.

Input data, d, is written at the address, a. If the write-through option is enabled, the input data is propagated through to the output bus, q, otherwise the output bus, q, remains stable.

If the word-write mask feature is implemented, via the generator, data on the data input bus is partitioned to the write mask bus, wen.Each wen pin has a distinct latched value, making each partition individually selectable.When the latched value of a write mask pin, wen, is low the corresponding data partition is selected, and its data is written to the memory location specified on the address bus.If the write mask pin, wen is high then no data is written for that partition.

If the write-through option is enabled, the write data is propagated to the output, q.If the write-through option is enabled the existing contents of the memory location is propagated to the output bus, q, otherwise the output bus, q, remains stable.

A standby mode is provided for periods of non-operation (cen=1). While in standby mode, address and data inputs are disabled; data stored in the memory is retained, but the memory cannot be accessed for reads or writes.

Memory normal mode is enabled (ret1n=1). In this mode the core and periphery power are both connected to the chip level power grid through Artigrid There is a power sequence when the memory is put from active to selective precharge and back to active. Selective precharge is available for all compilers except for the ROM. Before entering selective precharge, the memory must be in standby mode by setting cen=1, cen=1.

Extra Margin Adjustment pins provide the option of adding delays into internal timing pulses. There are 3 different EMA pins: ema, emaw, emas to control Read/Write internal timing pulses.

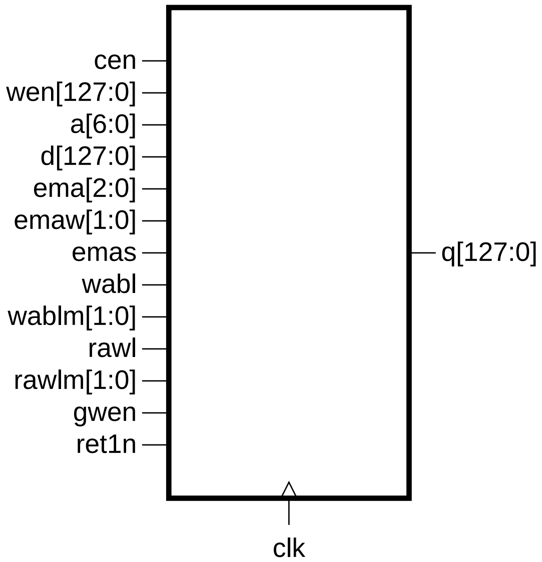
Refer to the user guide for a more detailed description of memory operation.

Physical Dimensions

Area Type	Width (μm)	Height (μm)	Area (μm^2)
Core	42.645	206.025	8785.94

All width, height, and area dimensions are in drawn dimensions. For shrink processes, this will be larger than the final silicon post-shrink dimensions.

Symbol

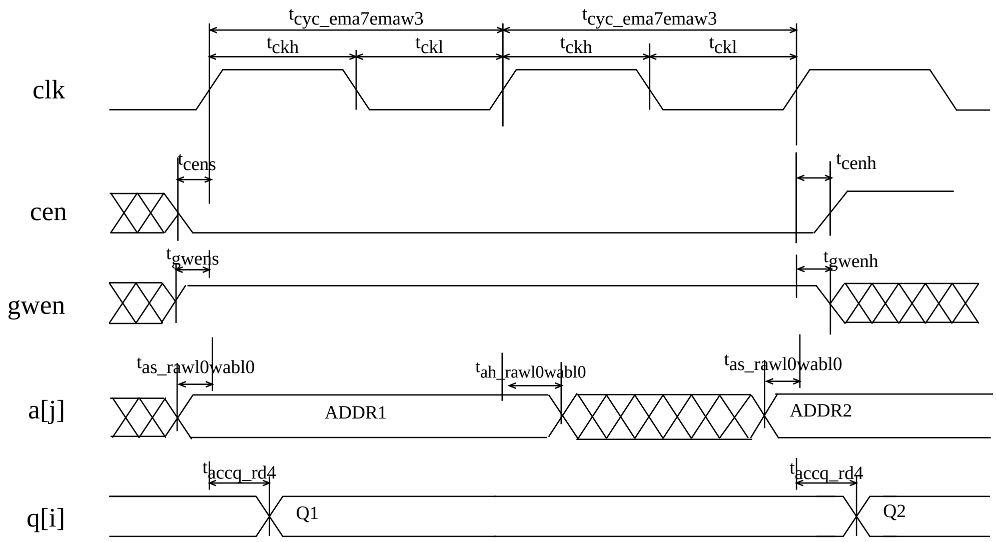


Pin Description

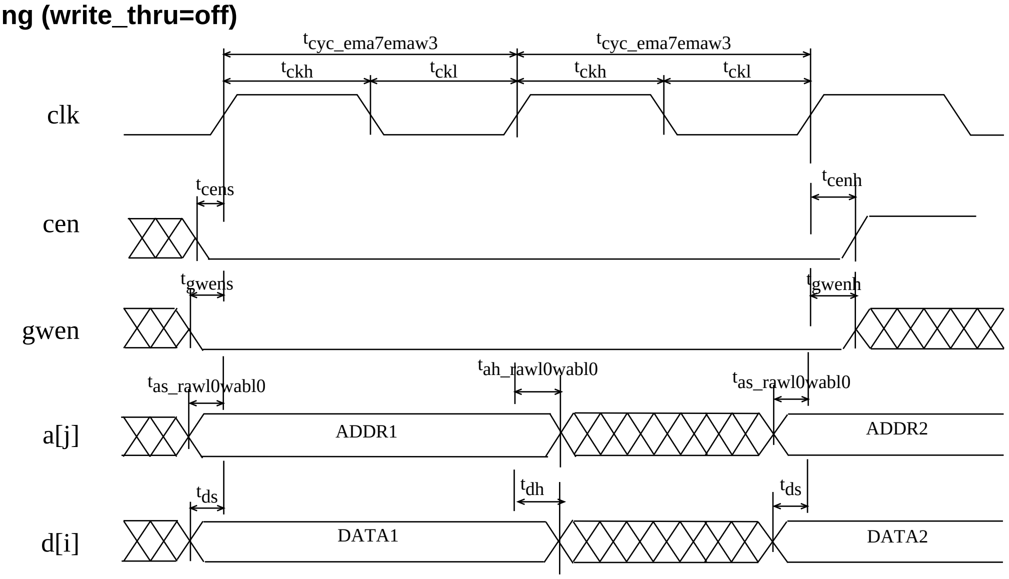
Pin	Description
a[6:0]	Address (a[0] = LSB)
d[127:0]	Data Input (d[0] = LSB)
clk	Clock
cen	Chip Enable (active low)
gwen	Write Enable (active low)
wen[127:0]	Write Enable (active low, wen[0] = LSB)
ema[2:0]	Extra Margin Adjustment (ema[0] = LSB)
emaw[1:0]	Write Extra Margin Adjustment (emaw[0] = LSB)
emas	Read Extra Margin Adjustment
ret1n	Retention Input (active low)
q[127:0]	Data Output (q[0] = LSB)

Read Cycle Timing

The retain timing arc is not shown in this diagram. Please refer to the User Guide for this compiler for a detailed timing diagram with the retain arc.



Write Cycle Timing (write_thru=off)



Default Timing for Cycle and Access (units = ns)

The timing tables shows delay values measured from 50% of supply to 50% of supply voltage. The output pins are loaded with the standard load of 0.035pF. Input pins are driven with a standard slew of 0.080ns from 10% to 90% of supply voltage.

The timing and power values are measured at input slew of 0.08ns on clock pin, 0.08ns on signal pins and output load 0.035pF.

Pin	Symbol	ffg Process 0.88V, -40°C	
		Min	Max
Delay clk to q ema=4[1,2]	t_{accq_rd4}	0.2017	0.3355
Min. Cycle clk ema=4 emaw=1 emas=0	$t_{cyc\_ema4emaw1}$	0.4617	
Min. High pulse width clk	t_{ckh}	0.1207	
Min. Low pulse width clk	t_{ckl}	0.0999	

[1] *Output delays and a load dependency (Kload) which is used to calculate:TotalDelay = FixedDelay + (Kload x Cload).*
[2] *Max access time is defined as the longest possible delay to valid output and min access time is defined as the shortest possible delay.*

Load Timing (units = ns/pF)

Pin	Symbol	ffg Process 0.88V, -40°C	
		Min	Max
q load factor	K_{load_q}		0.4316

[1] *The output load factor units are ns/pF.*

Setup and Hold Timing (units = ns)

Pin	Symbol	ffg Process 0.88V, -40°C	
		Min	Max
Setup Btw. clk and a rawl=1 wabl=1	$t_{as\_rawl1wabl1}$	555.0000	
Setup Btw. clk and a rawl=0 wabl=1	$t_{as\_rawl0wabl1}$	0.0787	

Setup and Hold Timing continued. (units = ns)

Pin	Symbol	ffg Process 0.88V, -40°C Min	Max
Setup Btw. clk and a rawl=1 wabl=0	$t_{as\_rawl1wabl0}$	555.0000	
Setup Btw. clk and a rawl=0 wabl=0	$t_{as\_rawl0wabl0}$	555.0000	
Hold Btw. clk and a rawl=1 wabl=1	$t_{ah\_rawl1wabl1}$	555.0000	
Hold Btw. clk and a rawl=0 wabl=1	$t_{ah\_rawl0wabl1}$	0.1018	
Hold Btw. clk and a rawl=1 wabl=0	$t_{ah\_rawl1wabl0}$	555.0000	
Hold Btw. clk and a rawl=0 wabl=0	$t_{ah\_rawl0wabl0}$	555.0000	
Setup Btw. clk and cen	t_{cens}	0.0801	
Hold Btw. clk and cen	t_{cenh}	0.0961	
Hold Btw. ret1n and cen	$t_{cenf\_ret1nfh}$	0.4597	
Hold Btw. ret1n and cen	$t_{cenf\_ret1nrh}$	0.2167	
Setup Btw. clk and wen	t_{wens}	0.0400	
Hold Btw. clk and wen	t_{wenh}	0.1599	
Setup Btw. clk and d	t_{ds}	0.0378	
Hold Btw. clk and d	t_{dh}	0.1688	
Setup Btw. clk and ema	t_{emas}	0.4479	
Hold Btw. clk and ema	t_{emah}	0.5728	
Setup Btw. clk and emaw	t_{emaws}	0.4479	
Hold Btw. clk and emaw	t_{emawh}	0.5728	
Setup Btw. clk and emas	t_{emass}	0.4479	
Hold Btw. clk and emas	t_{emash}	0.5728	
Setup Btw. clk and wabl	t_{wabls}	0.4479	
Hold Btw. clk and wabl	t_{wablh}	0.5507	
Setup Btw. clk and wablm	t_{wablms}	0.4479	
Hold Btw. clk and wablm	t_{wablmh}	0.5728	
Setup Btw. clk and rawl	t_{rawls}	0.4479	
Hold Btw. clk and rawl	t_{rawlh}	0.5507	
Setup Btw. clk and rawlm	t_{rawlms}	0.4479	
Hold Btw. clk and rawlm	t_{rawlmh}	0.5728	
Setup Btw. clk and gwen	t_{gwens}	0.0752	
Hold Btw. clk and gwen	t_{gwenh}	0.0985	
Hold Btw. cen and ret1n	t_{ret1nr_cenrh}	0.4589	
Hold Btw. cen and ret1n	t_{ret1nf_cenrh}	0.0417	

Cycle and Access Timing for Different Values of Extra Margin Adjustment (units = ns)

Pin	Symbol	ffg Process 0.88V, -40°C Min	Max
Delay clk to q ema=0[1,2]	t_{accq_rd0}	0.2017	0.2734
Delay clk to q ema=1[1,2]	t_{accq_rd1}	0.2017	0.2783
Delay clk to q ema=2[1,2]	t_{accq_rd2}	0.2017	0.2831
Delay clk to q ema=3[1,2]	t_{accq_rd3}	0.2017	0.2937
Delay clk to q ema=4[1,2]	t_{accq_rd4}	0.2017	0.3355
Delay clk to q ema=5[1,2]	t_{accq_rd5}	0.2017	0.3551
Delay clk to q ema=6[1,2]	t_{accq_rd6}	0.2017	0.3782
Delay clk to q ema=7[1,2]	t_{accq_rd7}	0.2017	0.3980
Min. Cycle clk ema=0 emaw=0 emas=0	$t_{cyc\_ema0emaw0}$	0.3823	
Min. Cycle clk ema=1 emaw=0 emas=0	$t_{cyc\_ema1emaw0}$	0.3871	
Min. Cycle clk ema=2 emaw=0 emas=0	$t_{cyc\_ema2emaw0}$	0.3918	
Min. Cycle clk ema=3 emaw=0 emas=0	$t_{cyc\_ema3emaw0}$	0.4021	
Min. Cycle clk ema=4 emaw=0 emas=0	$t_{cyc\_ema4emaw0}$	0.4428	
Min. Cycle clk ema=5 emaw=0 emas=0	$t_{cyc\_ema5emaw0}$	0.4618	
Min. Cycle clk ema=6 emaw=0 emas=0	$t_{cyc\_ema6emaw0}$	0.4842	
Min. Cycle clk ema=7 emaw=0 emas=0	$t_{cyc\_ema7emaw0}$	0.5035	
Min. Cycle clk ema=0 emaw=1 emas=0	$t_{cyc\_ema0emaw1}$	0.4013	
Min. Cycle clk ema=1 emaw=1 emas=0	$t_{cyc\_ema1emaw1}$	0.4061	
Min. Cycle clk ema=2 emaw=1 emas=0	$t_{cyc\_ema2emaw1}$	0.4107	
Min. Cycle clk ema=3 emaw=1 emas=0	$t_{cyc\_ema3emaw1}$	0.4210	

Cycle and Access Timing for Different Values of Extra Margin Adjustment continued (units = ns)

Pin	Symbol	ffg Process 0.88V, -40°C Min	Max
Min. Cycle clk ema=4 emaw=1 emas=0	$t_{cyc\_ema4emaw1}$	0.4617	
Min. Cycle clk ema=5 emaw=1 emas=0	$t_{cyc\_ema5emaw1}$	0.4808	
Min. Cycle clk ema=6 emaw=1 emas=0	$t_{cyc\_ema6emaw1}$	0.5032	
Min. Cycle clk ema=7 emaw=1 emas=0	$t_{cyc\_ema7emaw1}$	0.5224	
Min. Cycle clk ema=0 emaw=2 emas=0	$t_{cyc\_ema0emaw2}$	0.4249	
Min. Cycle clk ema=1 emaw=2 emas=0	$t_{cyc\_ema1emaw2}$	0.4297	
Min. Cycle clk ema=2 emaw=2 emas=0	$t_{cyc\_ema2emaw2}$	0.4344	
Min. Cycle clk ema=3 emaw=2 emas=0	$t_{cyc\_ema3emaw2}$	0.4447	
Min. Cycle clk ema=4 emaw=2 emas=0	$t_{cyc\_ema4emaw2}$	0.4854	
Min. Cycle clk ema=5 emaw=2 emas=0	$t_{cyc\_ema5emaw2}$	0.5044	
Min. Cycle clk ema=6 emaw=2 emas=0	$t_{cyc\_ema6emaw2}$	0.5268	
Min. Cycle clk ema=7 emaw=2 emas=0	$t_{cyc\_ema7emaw2}$	0.5461	
Min. Cycle clk ema=0 emaw=3 emas=0	$t_{cyc\_ema0emaw3}$	0.4433	
Min. Cycle clk ema=1 emaw=3 emas=0	$t_{cyc\_ema1emaw3}$	0.4481	
Min. Cycle clk ema=2 emaw=3 emas=0	$t_{cyc\_ema2emaw3}$	0.4528	
Min. Cycle clk ema=3 emaw=3 emas=0	$t_{cyc\_ema3emaw3}$	0.4631	
Min. Cycle clk ema=4 emaw=3 emas=0	$t_{cyc\_ema4emaw3}$	0.5037	
Min. Cycle clk ema=5 emaw=3 emas=0	$t_{cyc\_ema5emaw3}$	0.5228	
Min. Cycle clk ema=6 emaw=3 emas=0	$t_{cyc\_ema6emaw3}$	0.5452	
Min. Cycle clk ema=7 emaw=3 emas=0	$t_{cyc\_ema7emaw3}$	0.5644	

Cycle and Access Timing for Different Values of Extra Margin Adjustment continued (units = ns)

Pin	Symbol	ffg Process 0.88V, -40°C	
		Min	Max
High pulse width clk	t_{ckh}	0.1207	
Low pulse width clk	t_{ckl}	0.0999	

[1] *Output delays and a load dependency (Kload) which is used to calculate:TotalDelay = FixedDelay + (Kload x Cload).*
[2] *Max access time is defined as the longest possible delay to valid output and min access time is defined as the shortest possible delay.*

Pin Capacitance (units = fF)

Pin	ffg Process 0.88V, -40°C
clk	9.4360
cen	3.1971
wen	1.1481
a	4.1899
d	1.0914
ema	5.9463
emaw	3.2101
emas	4.3695
wabl	4.1899
wablm	4.1899
rawl	4.1899
rawlm	4.1899
gwen	3.2008
ret1n	6.1780

Current (units = mA)

Pin	ffg Process 0.88V, -40°C
Core Standby Chip Enable Curr.[3]	1.479e-03
Peri Standby Chip Enable Curr.[3]	1.677e-03
Core Standby std Curr.[3]	9.636e-04
Peri Standby std Curr.[3]	1.677e-03
Core Standby Selective Precharge Curr.[3]	9.512e-04
Peri Standby Selective Precharge Curr.[3]	1.556e-03
Core Standby Retention-1 Curr.[3]	9.512e-04

Current continued (units = mA)

Pin	ffg Process 0.88V, -40°C
Peri Standby Retention-1 Curr.[3]	2.340e-04
Core Read AC (ema=0) Curr.[1,4]	0.034309
Core Read AC (ema=1) Curr.[1,4]	0.034775
Core Read AC (ema=2) Curr.[1,4]	0.034779
Core Read AC (ema=3) Curr.[1,4]	0.034782
Core Read AC (ema=4) Curr.[1,4]	0.034786
Core Read AC (ema=5) Curr.[1,4]	0.034789
Core Read AC (ema=6) Curr.[1,4]	0.034792
Core Read AC (ema=7) Curr.[1,4]	0.034795
Peri Read AC (ema=0) Curr.[1,4]	1.406076
Peri Read AC (ema=1) Curr.[1,4]	1.420238
Peri Read AC (ema=2) Curr.[1,4]	1.427399
Peri Read AC (ema=3) Curr.[1,4]	1.445085
Peri Read AC (ema=4) Curr.[1,4]	1.502417
Peri Read AC (ema=5) Curr.[1,4]	1.536765
Peri Read AC (ema=6) Curr.[1,4]	1.572732
Peri Read AC (ema=7) Curr.[1,4]	1.595902
Core avgOfReadWrite AC (ema=0) Curr.[1,4]	0.035998
Core avgOfReadWrite AC (ema=1) Curr.[1,4]	0.036488
Core avgOfReadWrite AC (ema=2) Curr.[1,4]	0.036491
Core avgOfReadWrite AC (ema=3) Curr.[1,4]	0.036495
Core avgOfReadWrite AC (ema=4) Curr.[1,4]	0.036499
Core avgOfReadWrite AC (ema=5) Curr.[1,4]	0.036502
Core avgOfReadWrite AC (ema=6) Curr.[1,4]	0.036505
Core avgOfReadWrite AC (ema=7) Curr.[1,4]	0.036508
Peri avgOfReadWrite AC (ema=0) Curr.[1,4]	2.235191
Peri avgOfReadWrite AC (ema=1) Curr.[1,4]	2.249353
Peri avgOfReadWrite AC (ema=2) Curr.[1,4]	2.256637
Peri avgOfReadWrite AC (ema=3) Curr.[1,4]	2.274204
Peri avgOfReadWrite AC (ema=4) Curr.[1,4]	2.331535
Peri avgOfReadWrite AC (ema=5) Curr.[1,4]	2.365881
Peri avgOfReadWrite AC (ema=6) Curr.[1,4]	2.401849
Peri avgOfReadWrite AC (ema=7) Curr.[1,4]	2.425019
Core Write AC (ema=0 emaw=1) Curr.[1,4]	0.037687
Core Write AC (ema=1 emaw=1) Curr.[1,4]	0.0382
Core Write AC (ema=2 emaw=1) Curr.[1,4]	0.038204
Core Write AC (ema=3 emaw=1) Curr.[1,4]	0.038208
Core Write AC (ema=4 emaw=1) Curr.[1,4]	0.038212
Core Write AC (ema=5 emaw=1) Curr.[1,4]	0.038215
Core Write AC (ema=6 emaw=1) Curr.[1,4]	0.038218
Core Write AC (ema=7 emaw=1) Curr.[1,4]	0.038222
Peri Write AC (ema=0 emaw=1) Curr.[1,4]	3.064306

Current continued (units = mA)

Pin	ffg Process 0.88V, -40°C
Peri Write AC (ema=1 emaw=1) Curr.[1,4]	3.078469
Peri Write AC (ema=2 emaw=1) Curr.[1,4]	3.085874
Peri Write AC (ema=3 emaw=1) Curr.[1,4]	3.103323
Peri Write AC (ema=4 emaw=1) Curr.[1,4]	3.160653
Peri Write AC (ema=5 emaw=1) Curr.[1,4]	3.194997
Peri Write AC (ema=6 emaw=1) Curr.[1,4]	3.230966
Peri Write AC (ema=7 emaw=1) Curr.[1,4]	3.254135
Core Deselect (icc_c_desel) Curr.[2,4]	0.000e+00
Peri Deselect (icc_p_desel) Curr.[2,4]	0.070075
Core Peak (icc_c_peak) Curr.	10.334391
Peri Peak (icc_p_peak) Curr.	110.915016
Core Inrush (icc_c_inrush) Curr.	2.277195
Peri Inrush (icc_p_inrush) Curr.	46.200867

[1] *The AC current value assumes 50% read and write operations, where 50% addresses and 50% of input and output pins switch at the user defined frequency of 500MHz and user defined clock activity_factor of 50%. It is assumed that ema pins do not switch.*
[2] *The deselected current assumes the memory is deselected, 50% addresses switch, and 50% of input pins switch at the user defined frequency of 500MHz. The logic switching component of deselected power becomes negligbly small if the input pins are held stable by externally controlling these signals with chip select. It is assumed that ema pins do not switch.*
[3] *The standby current value is independent of frequency and assumes all inputs and outputs are stable.*
[4] *The leakage current component is not included in this value.*
[5] *Clock activity factor will affect total current.*

Clock Noise Limit (Time-units = ns, Voltage-units = V)

Symbol	ffg Process 0.88V, -40°C	
	Pulse Width	Voltage
clk	0.0603	0.1760

The clock noise limit is the maximum voltage allowed (for the indicated pulse width) that does not cause an unintentional memory cycle or other memory failure.

Supply Noise Limit (units = V)

Pin	ffg Process 0.88V, -40°C
Power	0.0880
Ground	0.0880

The power and ground noise limit is the maximum supply voltage transition that is allowed without causing a memory failure.